VITyarthi | Open Source Software

The Open Source Audit

A Capstone Project for OSS NGMC Course

Course: Open Source Software | Unit Coverage: 1 – 5

Max Marks: 100

Student Name	Registration Number
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Slot	**Date of Submission**
A24	31-03-2026

1. Introduction

The world of open-source software has revolutionized the modern digital world with its power of collaboration, transparency, and innovation. One such example of open-source software is VLC Media Player, which is used for multimedia purposes.

The VLC Media Player originated in 1996 as a project undertaken by students of École Centrale Paris in France. Initially, it was intended to be used for video streaming purposes across the university campus network. In those days, it was difficult to find multimedia software that could handle multiple formats. Students wanted free software that could stream educational materials across the campus network.

This idea became the starting point for creating something called VLC.

2. Choosing Your Software

Pick one open-source project from the list below, or propose your own (subject to instructor approval). Your choice will be the subject of your entire report and all five shell scripts.

Approved software options

Software	Category	License	Why it's interesting
Linux Kernel	Operating System	`GPL v2`	*The foundation everything else runs on*
Apache HTTP Server	Web Server	`Apache 2.0`	*Powers ~30% of all websites globally*
MySQL	Database	`GPL v2 / Commercial`	*A license with a dual-model story to tell*
Firefox	Web Browser	`MPL 2.0`	*A nonprofit fighting for an open web*
VLC Media Player	Multimedia	`LGPL / GPL`	*Plays anything — built by students in Paris*
LibreOffice	Office Suite	`MPL 2.0`	*Born from a community fork — a real lesson*
Python	Programming Language	`PSF License`	*A language shaped entirely by community*
Git	Version Control	`GPL v2`	*The tool Linus built when proprietary failed him*

Your choice	Write your chosen software here before submission: ~~VLC Media Player~~

3. Project Report Structure

Part A — Origin and Philosophy (Units 1 & 2)

A1. The problem VLC was created to solve

VLC Media Player is arguably one of the most interesting stories in the development of open-source software. It was initially developed in **1996 by students at École Centrale Paris, France**, as an academic project. The main problem they wanted to solve was to **enable video streaming on the university campus network**.

At the time, multimedia software was highly limited. Most media players could only support a few file formats, needed paid codecs, or functioned only in specific proprietary environments. This made it difficult for students and universities to stream educational videos and other learning content smoothly across different systems.

The students who developed the software wanted something much better. Instead of building software only for personal use, they wanted to make it available for everyone to use, improve, and expand. This reflects the core philosophy of open source — **solve a problem once and allow the world to benefit from it**.

What started as a solution for campus network streaming gradually evolved into **VLC Media Player**, which today is used by millions of people worldwide for video playback, streaming, and media conversion. Its journey from a university project to a globally trusted application clearly shows the power of collaborative software development.

A2. The philosophy behind sharing VLC openly

The VLC team recognized that access to multimedia content should not be restricted through company lock-ins or proprietary systems. Their decision to distribute VLC under an open-source license showed the belief that software should **empower users instead of limiting them**.

This belief is closely aligned with the general principles of free and open-source software:

- Transparency
- Collaboration
- Freedom to improve the code
- Freedom to redistribute the code

Because the source code is openly available, programmers around the world have continuously improved VLC by adding support for new codecs, optimizing performance, improving subtitles, and extending streaming capabilities.

This is the reason VLC Media Player can play almost every audio and video format today. The open-source philosophy has transformed it from a simple media player into a universal multimedia solution supported by a global community.

A3. Ethical reflection — Why is open source important in this case?

The case of VLC Media Player is especially important in the context of open source because access to multimedia content has become an essential need in the digital age. Educational videos, online lectures, communication, and entertainment all depend on reliable media tools.

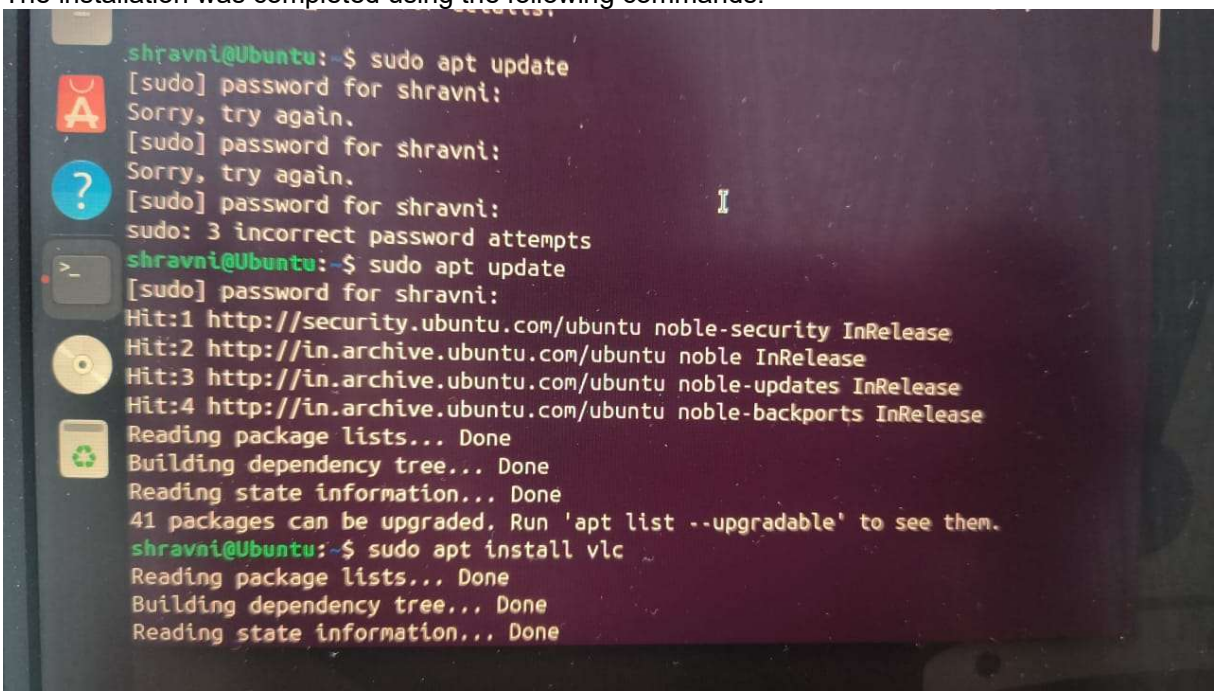
If the tools required to access such content remain proprietary and expensive, many people may be denied access to important information. This can create a digital divide between those who can afford technology and those who cannot.

The decision to distribute VLC through an open-source license ensures **equal access, trust, and transparency**. Since the source code is available, anyone can understand what the software is doing, verify its safety, and improve it according to their needs.

This is ethical because fairness is achieved by making the technology available to students, programmers, and organizations irrespective of their economic background. VLC Media Player is therefore not just a useful tool, but also an embodiment of the ethical belief that **knowledge and digital resources should remain accessible to all**.

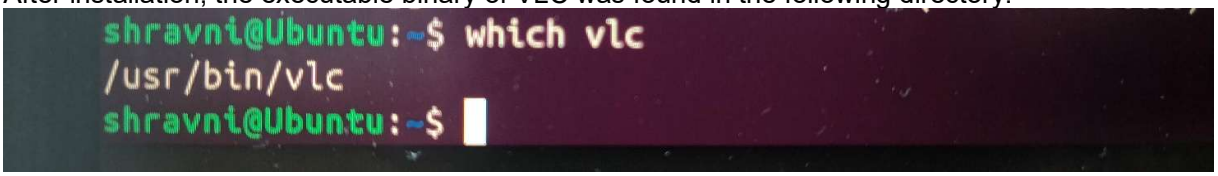
Part B — Linux Footprint (Unit 2)

After setting up Ubuntu Linux in a VirtualBox environment, VLC Media Player was installed using the **APT package manager**, which is the default package management system for Debian-based Linux distributions.
The installation was completed using the following commands:



This method is simple and secure because the software is downloaded directly from Ubuntu's official repositories. One of the major strengths of open-source software is that installation and updates are integrated into the operating system's package management system.

After installation, the executable binary of VLC was found in the following directory:



This directory stores user-level executable binaries that can be launched directly from the terminal.
The configuration files of VLC are stored inside the user's home directory:

This folder contains:

- user preferences
- playback settings
- interface customizations
- cache references
- recently played media details

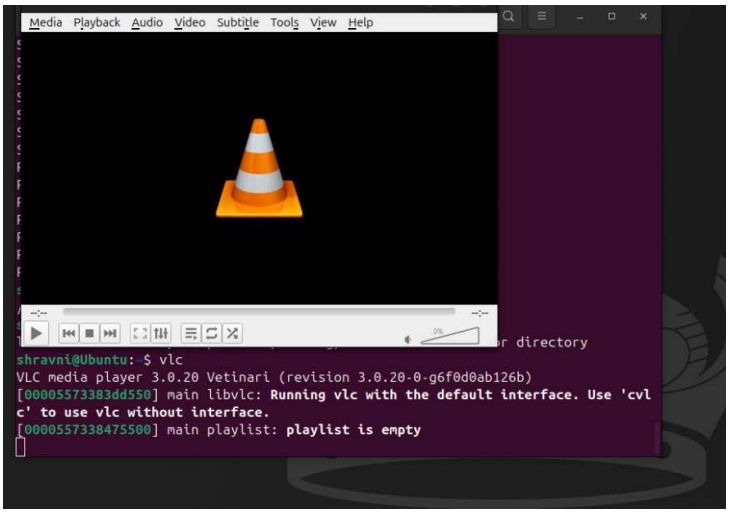
This structure is important because Linux separates executable files from user-specific configuration data, improving system organization and maintainability.

User, permissions, and process behavior

Unlike server software, VLC does not run as a dedicated Linux service account. Instead, it runs under the **currently logged-in user account**.
This is important for security because:

- VLC only accesses files the user is allowed to access
- system-wide changes require root permission
- personal settings remain isolated per user

The process can be started using:



or through the Ubuntu application launcher.
Since VLC is mainly a desktop multimedia application, it is not usually managed using systemctl like background services such as Apache or MySQL. Instead, it starts as a foreground GUI process.

Update model and security patches
VLC receives updates through Ubuntu repositories. The open-source community, Ubuntu maintainers, and VideoLAN contributors regularly release:

- bug fixes
- security patches
- codec updates
- performance improvements

These updates can be installed using:
sudo apt upgrade
This community-driven patch model is one of the biggest strengths of open-source software because vulnerabilities can be identified and fixed quickly by contributors worldwide.

Part C — The FOSS Ecosystem (Units 3 & 4)

The VLC Media Player is part of a vast free and open-source software landscape. It uses critical open-source software libraries such as FFmpeg, x264, and Qt. The libraries enable the VLC Media Player to play various audio and video formats.
The VLC Media Player has also led to the development of various streaming tools, IPTV players, and educational video sites. The VLC Media Player is highly extensible with the ability to add various plugins such as subtitle tools and codec plugins.
The VLC Media Player is maintained by the VideoLAN community. The community uses various tools such as Git repositories and forums. The open-source software landscape demonstrates that open-source software development is all about collaboration. The VLC Media Player depends on and contributes to the development of various open-source software.

Part D — Open Source vs Proprietary (Critical Analysis)

Dimension	Open Source(VLC Media Player)	Proprietary Alternative(Windows Media Player)
Cost	Free to download and use on all platforms	Included with Windows, not separately free on other platforms
Security (who can audit the code?)	Included with Windows, not separately free on other platforms	Source code is closed and only Microsoft can audit it
Support and reliability	Supported by Video LAN community, forums, and regular updates	Official Microsoft support, stable in Windows ecosystem
Freedom to modify	Users and developers can modify and redistribute under GPL	Users cannot modify source code
Community vs corporate control	Controlled by global open-source community	Fully controlled by Microsoft
Your overall verdict	Better for flexibility, format support, and transparency	Better only for default Windows integration

After comparing the features of VLC Media Player and Windows Media Player, it is clear that VLC Media Player is the better option for real-world application. The most important features of VLC Media Player are that it supports multiple formats, it is transparent, and it is modifiable. Since the source code is open source, it is safe to use and can be modified according to the needs of the user. For the real-world application of the media player, I recommend the VLC Media Player. I am also interested in contributing to the project because it is the true spirit of open-source development.

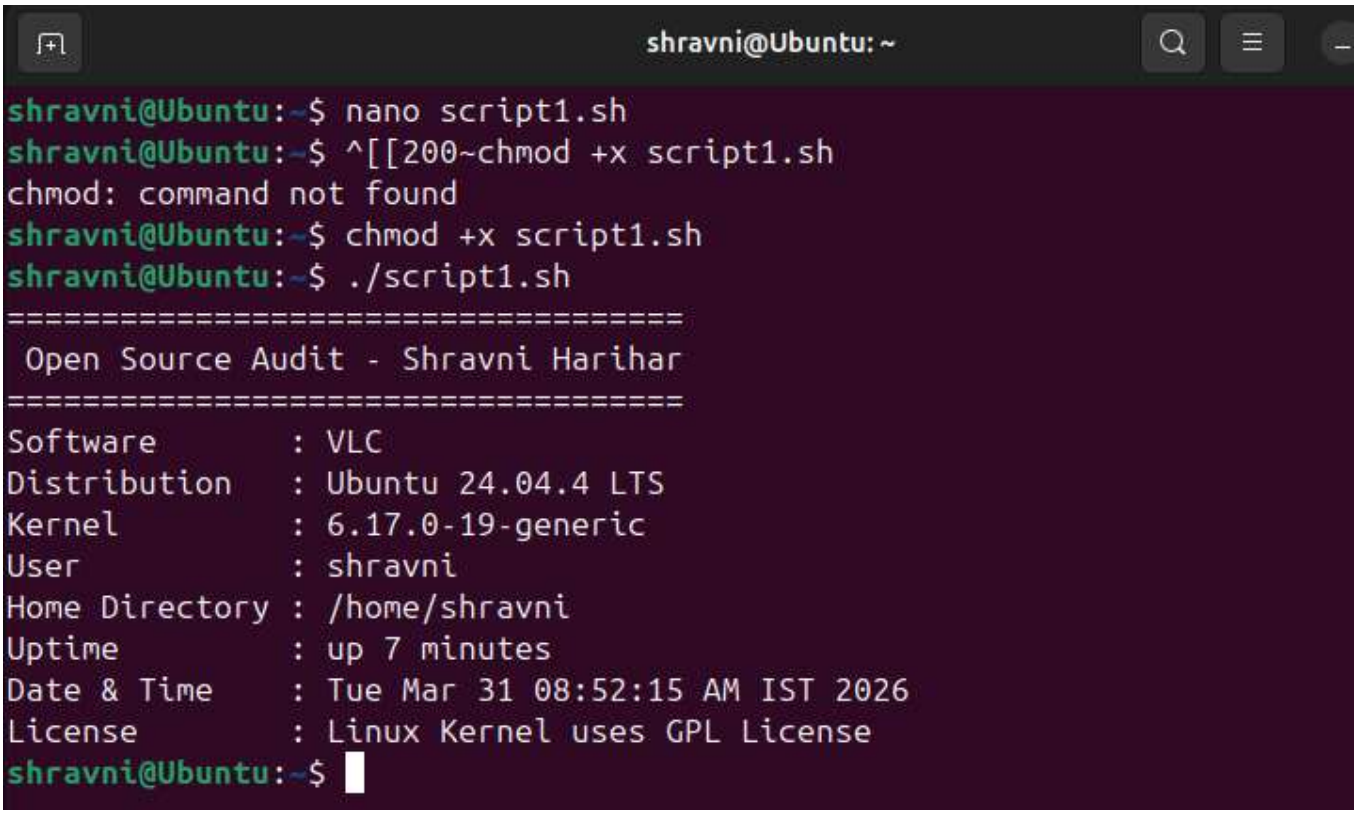
4. Shell Script Tasks

You will write five shell scripts. Each script must be submitted as a plain .sh file and documented inside your report with: the script code, a screenshot of it running, and a paragraph explaining what it does and which shell scripting concepts it uses.

Important	Scripts must run on a real Linux system. They must include comments explaining each section. Uncommented scripts will lose marks.

Script 1 — System Identity Report (Unit 1 & 2)

This script is designed as a **System Identity Report** for the Linux environment used in the project. It collects and displays important system details such as the Linux distribution name, kernel version, logged-in user, home directory, uptime, and current date and time. The script uses **variables, command substitution ($()), echo statements, and basic output formatting** to present the information in a structured way. It demonstrates how shell scripting can automate system information retrieval and improve readability of technical details.



Script 2 — FOSS Package Inspector (Unit 2)

This script works as a **FOSS Package Inspector** for VLC Media Player. It first checks whether the VLC package is installed on the Ubuntu system using an **if-then-else condition** with the dpkg command. If the package is found, the script extracts useful details such as version, maintainer, and description. A **case statement** is then used to print a short philosophy note based on the package name. This script demonstrates the use of **conditional statements, package inspection commands, pipes, grep, and case control structures** in shell scripting.

```
shravni@Ubuntu:~$ nano script2.sh
shravni@Ubuntu:~$ chmod +x script2.sh
shravni@Ubuntu:~$ ./script2.sh
vlc is installed.
Maintainer: Ubuntu Developers <ubuntu-devel-discuss@lists.ubuntu.com>
Version: 3.0.20-3build6
Description: multimedia player and streamer
Original-Maintainer: Debian Multimedia Maintainers <debian-multimedia@lists.debian.org>
VLC: open-source player that supports almost every media format
```

Script 3 — Disk and Permission Auditor (Unit 2)

This script performs a **Disk and Permission Audit** of important Linux directories. Using a **for loop**, it iterates through directories such as /etc, /var/log, /home, /usr/bin, and /tmp. For each directory, it checks whether it exists and then reports its permissions, ownership, and size. The script also verifies the VLC configuration directory inside the user's home folder. This demonstrates the use of **loops, directory testing, command substitution, ls -ld, du, awk, and cut commands** in shell scripting.

```
shravni@Ubuntu:~$ nano script3.sh
shravni@Ubuntu:~$ chmod +x script3.sh
shravni@Ubuntu:~$ ./script3.sh
Directory Audit Report
--------------------------------
/etc => Permissions: drwxr-xr-x root root | Size: 12M
/var/log => Permissions: drwxrwxr-x root syslog | Size: 40M
/home => Permissions: drwxr-xr-x root root | Size: 368M
/usr/bin => Permissions: drwxr-xr-x root root | Size: 217M
/tmp => Permissions: drwxrwxrwt root root | Size: 80K
VLC Config => drwx------ shravni shravni
```

Script 4 — Log File Analyzer (Units 2 & 5)

This script is a **Log File Analyzer** that reads a log file line by line using a **while-read loop**. It searches for a user-specified keyword such as error and counts how many times that keyword appears in the file. At the end, it displays the total count and shows the last matching lines. This script demonstrates the use of **command-line arguments ($1), default values, counter variables, while loops, if conditions, grep, and tail commands**. A sample log file was used to clearly show the working of the script.

```
shravni@Ubuntu:~$ nano sample.log
shravni@Ubuntu:~$ ./script4.sh sample.log error
Keyword 'error' found 3 times in sample.log
ERROR: VLC codec missing
ERROR: failed to open media file
ERROR: subtitle file not found
shravni@Ubuntu:~$
```

Script 5 — The Open Source Manifesto Generator (Unit 5)

This script acts as an **Open Source Manifesto Generator**. It interactively asks the user three questions related to open-source philosophy using the read command. Based on the answers, it generates a personalized manifesto paragraph and saves it into a text file with the current username. This script demonstrates the use of **interactive input, variables, string concatenation, date formatting, file writing using redirection (>), and output display using cat**. It creatively connects shell scripting with the philosophy of open-source software.

```
shravni@Ubuntu:~$ nano script5.sh
shravni@Ubuntu:~$ chmod +x script5.sh
shravni@Ubuntu:~$ ./script5.sh

Answer three questions to generate your manifesto.

1. Name one open-source tool you use every day: vlc
2. In one word, what does freedom mean to you? freedom
3. Name one friend you would build and share freely: build

Manifesto saved to manifesto_shravni.txt
On 31 March 2026, I pledge tools like vlc represent freedom. I would love to build and freely share build with the wor
d.
shravni@Ubuntu:~$
```